PREMIUM

Full-Length Mock Test

Mock 02

Engineering Sciences

GATE XE

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | **Moderate 33%** | **Hard 47%**

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Engineering Mathematics — Complex Variables]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.2 [1 Mark] [EASY] [Solid Mechanics — Thin Cylinders]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.3 [1 Mark] [EASY] [Thermodynamics — Psychrometrics]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.4 [1 Mark] [EASY] [Material Science — Ceramics]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.5 [1 Mark] [EASY] [Engineering Mathematics — Calculus]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.6 [1 Mark] [EASY] [Solid Mechanics — Bending]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.7 [1 Mark] [EASY] [Thermodynamics — Laws]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$

B) $(1/2)MR^2$

C) $(2/3)MR^2$

D) $(3/5)MR^2$

Q.8 [1 Mark] [EASY] [Material Science — Crystal Structures]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.9 [1 Mark] [EASY] [Engineering Mathematics — Calculus]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.10 [1 Mark] [EASY] [Solid Mechanics — Thin Cylinders]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.11 [1 Mark] [EASY] [Thermodynamics — Refrigeration]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.12 [1 Mark] [EASY] [Material Science — Ceramics]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.13 [2 Marks] [MODERATE] [Engineering Mathematics — Calculus]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.14 [2 Marks] [MODERATE] [Solid Mechanics — Thin Cylinders]

Entropy change for reversible adiabatic process:

A) Zero

B) Maximum

C) Minimum

D) Positive

Q.15 [2 Marks] [MODERATE] [Thermodynamics — Power Cycles]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.16 [2 Marks] [MODERATE] [Material Science — Corrosion]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.17 [2 Marks] [MODERATE] [Engineering Mathematics — ODE]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.18 [2 Marks] [MODERATE] [Solid Mechanics — Thin Cylinders]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.19 [2 Marks] [MODERATE] [Thermodynamics — Entropy]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.20 [2 Marks] [MODERATE] [Material Science — Ceramics]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.21 [2 Marks] [MODERATE] [Engineering Mathematics — ODE]

Rankine cycle is used in:

A) Gas turbines

B) Steam power plants

C) Diesel engines

D) Jet engines

Q.22 [2 Marks] [MODERATE] [Solid Mechanics — Stress-Strain]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.23 [2 Marks] [MODERATE] [Thermodynamics — Psychrometrics]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.24 [2 Marks] [MODERATE] [Material Science — Crystal Structures]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.25 [2 Marks] [MODERATE] [Engineering Mathematics — Calculus]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.26 [2 Marks] [MODERATE] [Solid Mechanics — Thin Cylinders]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.27 [2 Marks] [MODERATE] [Thermodynamics — Power Cycles]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.28 [2 Marks] [MODERATE] [Material Science — Phase Diagrams]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$

B) $(1/2)MR^2$

C) (2/3)MR^2

D) $(3/5)MR^2$

Q.29 [2 Marks] [MODERATE] [Engineering Mathematics — Probability]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.30 [2 Marks] [MODERATE] [Solid Mechanics — Thin Cylinders]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.31 [2 Marks] [MODERATE] [Thermodynamics — Refrigeration]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.32 [2 Marks] [MODERATE] [Material Science — Phase Diagrams]

Entropy change for reversible adiabatic process:

A) Zero
B) Maximum
C) Minimum
D) Positive

Q.33 [2 Marks] [HARD] [Engineering Mathematics — Calculus]

Rankine cycle is used in:

A) Gas turbines
B) Steam power plants
C) Diesel engines
D) Jet engines

Q.34 [2 Marks] [HARD] [Solid Mechanics — Thin Cylinders]

Moment of inertia of solid sphere about diameter:

A) $(2/5)MR^2$
B) $(1/2)MR^2$
C) $(2/3)MR^2$
D) $(3/5)MR^2$

Q.35 [2 Marks] [HARD] [Thermodynamics — Refrigeration]

Entropy change for reversible adiabatic process:

A) Zero

B) Maximum

C) Minimum

D) Positive

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Material Science — Ceramics]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Engineering Mathematics — Complex Variables]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Solid Mechanics — Thin Cylinders]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Thermodynamics — Laws]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Material Science — Ceramics]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Thermodynamics — Psychrometrics]

Solid sphere I about diameter:

Answer: ______________________

Q.52 [2 Marks] [HARD] [Material Science — Corrosion]

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: ______________________

Q.53 [2 Marks] [HARD] [Engineering Mathematics — Complex Variables]

Solid sphere I about diameter:

Answer: ______________________

Q.54 [2 Marks] [HARD] [Solid Mechanics — Bending]

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: ______________________

Q.55 [2 Marks] [HARD] [Thermodynamics — Psychrometrics]

Solid sphere I about diameter:

Answer: ______________________

Q.56 [2 Marks] [HARD] [Material Science — Ceramics]

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: ______________________

Q.57 [2 Marks] [HARD] [Engineering Mathematics — Probability]

Solid sphere I about diameter:

Answer: ______________________

Q.58 [2 Marks] [HARD] [Solid Mechanics — Pressure Vessels]

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: ________________________

Q.59 [2 Marks] [HARD] [Thermodynamics — Psychrometrics]

Solid sphere I about diameter:

Answer: ___________________________

Q.60 [2 Marks] [HARD] [Material Science — Crystal Structures]

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: ___________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	A	**Q.2:**	A	**Q.3:**	B
Q.4:	A	**Q.5:**	A	**Q.6:**	B
Q.7:	A	**Q.8:**	A	**Q.9:**	B
Q.10:	A	**Q.11:**	A	**Q.12:**	B
Q.13:	A	**Q.14:**	A	**Q.15:**	B
Q.16:	A	**Q.17:**	A	**Q.18:**	B
Q.19:	A	**Q.20:**	A	**Q.21:**	B
Q.22:	A	**Q.23:**	A	**Q.24:**	B
Q.25:	A	**Q.26:**	A	**Q.27:**	B
Q.28:	A	**Q.29:**	A	**Q.30:**	B
Q.31:	A	**Q.32:**	A	**Q.33:**	B
Q.34:	A	**Q.35:**	A	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	A
Q.52:	50	**Q.53:**	A	**Q.54:**	50
Q.55:	A	**Q.56:**	50	**Q.57:**	A
Q.58:	50	**Q.59:**	A	**Q.60:**	50

Detailed Solutions

Question 1
[Engineering Mathematics — Complex Variables]

Moment of inertia of solid sphere about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere. Hollow sphere: (2/3)MR^2.

Question 2
[Solid Mechanics — Thin Cylinders]

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 3
[Thermodynamics — Psychrometrics]

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 4
[Material Science — Ceramics]

Moment of inertia of solid sphere about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere. Hollow sphere: (2/3)MR^2.

Question 5
[Engineering Mathematics — Calculus]

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 6
[Solid Mechanics — Bending]

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 7
[Thermodynamics — Laws]

Moment of inertia of solid sphere about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere. Hollow sphere: (2/3)MR^2.

Question 8
[Material Science — Crystal Structures]

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 9
[Engineering Mathematics — Calculus]

Rankine cycle is used in:

Answer:

(B)

Rankine cycle is the basis for steam power plant operation.

Question 10 *[Solid Mechanics — Thin Cylinders]*

Moment of inertia of solid sphere about diameter:

Answer: (A)

$I = (2/5)MR^2$ for solid sphere. Hollow sphere: $(2/3)MR^2$.

Question 11 *[Thermodynamics — Refrigeration]*

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 12 *[Material Science — Ceramics]*

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 13 *[Engineering Mathematics — Calculus]*

Moment of inertia of solid sphere about diameter:

Answer: (A)

$I = (2/5)MR^2$ for solid sphere. Hollow sphere: $(2/3)MR^2$.

Question 14 *[Solid Mechanics — Thin Cylinders]*

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 15 *[Thermodynamics — Power Cycles]*

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 16 *[Material Science — Corrosion]*

Moment of inertia of solid sphere about diameter:

Answer: (A)

$I = (2/5)MR^2$ for solid sphere. Hollow sphere: $(2/3)MR^2$.

Question 17 *[Engineering Mathematics — ODE]*

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 18 *[Solid Mechanics — Thin Cylinders]*

Rankine cycle is used in:

Answer:

(B)

Rankine cycle is the basis for steam power plant operation.

Question 19 *[Thermodynamics — Entropy]*

Moment of inertia of solid sphere about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere. Hollow sphere: (2/3)MR^2.

Question 20 *[Material Science — Ceramics]*

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 21 *[Engineering Mathematics — ODE]*

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 22 *[Solid Mechanics — Stress-Strain]*

Moment of inertia of solid sphere about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere. Hollow sphere: (2/3)MR^2.

Question 23 *[Thermodynamics — Psychrometrics]*

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 24 *[Material Science — Crystal Structures]*

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 25 *[Engineering Mathematics — Calculus]*

Moment of inertia of solid sphere about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere. Hollow sphere: (2/3)MR^2.

Question 26 *[Solid Mechanics — Thin Cylinders]*

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 27 *[Thermodynamics — Power Cycles]*

Rankine cycle is used in:

Answer:

(B)

Rankine cycle is the basis for steam power plant operation.

Question 28
[Material Science — Phase Diagrams]

Moment of inertia of solid sphere about diameter:

Answer: (A)

$I = (2/5)MR^2$ for solid sphere. Hollow sphere: $(2/3)MR^2$.

Question 29
[Engineering Mathematics — Probability]

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 30
[Solid Mechanics — Thin Cylinders]

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 31
[Thermodynamics — Refrigeration]

Moment of inertia of solid sphere about diameter:

Answer: (A)

$I = (2/5)MR^2$ for solid sphere. Hollow sphere: $(2/3)MR^2$.

Question 32
[Material Science — Phase Diagrams]

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 33
[Engineering Mathematics — Calculus]

Rankine cycle is used in:

Answer: (B)

Rankine cycle is the basis for steam power plant operation.

Question 34
[Solid Mechanics — Thin Cylinders]

Moment of inertia of solid sphere about diameter:

Answer: (A)

$I = (2/5)MR^2$ for solid sphere. Hollow sphere: $(2/3)MR^2$.

Question 35
[Thermodynamics — Refrigeration]

Entropy change for reversible adiabatic process:

Answer: (A)

Isentropic (reversible adiabatic): 9E2 Ò à

Question 36
[Material Science — Ceramics]

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Engineering Mathematics — Complex Variables]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 *[Solid Mechanics — Thin Cylinders]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 *[Thermodynamics — Laws]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 *[Material Science — Ceramics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 *[Engineering Mathematics — Complex Variables]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 *[Solid Mechanics — Stress-Strain]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 *[Thermodynamics — Psychrometrics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 *[Material Science — Ceramics]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 *[Engineering Mathematics — Linear Algebra]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Solid Mechanics — Thin Cylinders]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Thermodynamics — Entropy]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Material Science — Crystal Structures]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Engineering Mathematics — Calculus]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Solid Mechanics — Torsion]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Thermodynamics — Psychrometrics]*

Solid sphere I about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere about diameter.

Question 52 *[Material Science — Corrosion]*

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: (50)

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Question 53 *[Engineering Mathematics — Complex Variables]*

Solid sphere I about diameter:

Answer: (A)

I = (2/5)MR^2 for solid sphere about diameter.

Question 54 *[Solid Mechanics — Bending]*

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer:

(50)

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Question 55 *[Thermodynamics — Psychrometrics]*

Solid sphere I about diameter:

Answer: (A)

I = (2/5)MR² for solid sphere about diameter.

Question 56 *[Material Science — Ceramics]*

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: (50)

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Question 57 *[Engineering Mathematics — Probability]*

Solid sphere I about diameter:

Answer: (A)

I = (2/5)MR² for solid sphere about diameter.

Question 58 *[Solid Mechanics — Pressure Vessels]*

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: (50)

;r Ò Ò 3 óc Ò ãR Ò S Rà

Question 59 *[Thermodynamics — Psychrometrics]*

Solid sphere I about diameter:

Answer: (A)

I = (2/5)MR² for solid sphere about diameter.

Question 60 *[Material Science — Crystal Structures]*

Rankine cycle efficiency at T_h=600K, T_c=300K (Carnot) = ? (%)

Answer: (50)

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